A First Course in Causal Inference Ch. 11

Kodai Minesaki

Faculty of Engineering, The University of Tokyo

November 11, 2026

Ch. 11 The Central Role of the Propensity Score in Observational Studies for Causal Effects

Ch. 11 The Central Role of the Propensity Score in Observational Studies for Causal Effects

- Rosenbaum and Rubin (1983) proposed the key concept *propensity score*.
- Under the i.i.d. sampling assumption, we have four random variables associated with each unit: $\{X, Z, Y(1), Y(0)\}$.
- We can factorize the joint distribution as

$$\Pr\{X, Z, Y(1), Y(0)\} = \Pr\{X\} \cdot \Pr\{Y(1), Y(0) \mid X\} \cdot \Pr\{Z \mid X, Y(1), Y(0)\}.$$

- We do not want to model the covariates becasue they are background information happening before the treatment and outcome.
- If we want to go beyond the outcome model, then we must focus on the treatment assignment mechanism, which leads to the definition of the propensity score.

Definition 11.1 (propensity score)

Define

$$e(X, Y(1), Y(0)) = \Pr\{Z = 1 \mid X, Y(1), Y(0)\}$$

as the propensity score. Under strong ignorability, we have

$$e(X, Y(1), Y(0)) = \Pr\{Z = 1 \mid X, Y(1), Y(0)\} = \Pr(Z = 1 \mid X),$$

so the propensity score reduces to

$$e(X) = \Pr\{Z = 1 \mid X\},$$

the conditional probability of receriving the treatment given the observed covariates.

11.1 The propensity score as a dimension reduction tool

Theorem 11.1

$Z \perp\!\!\!\perp \{Y(1), Y(0)\} \mid X \Rightarrow Z \perp\!\!\!\perp \{Y(1), Y(0) \mid e(X)\}$.

- Thm. 11.1 implies that conditioning on the propensity score $e(X)$ removes all confunding induced by covariates X.
- The original covariates X can be high-dimensional but $e(X)$ is one-dimensional scalar variable bounded between 0 and 1.
- Therefore, the propensity score reduces the dimension of the original covariates but still maintains the ignorability: a *dimensional reduction tool*.

Proof.

We need to show

$$\Pr\{Z = 1 \mid Y(1), Y(0), e(X)\} = \Pr\{Z = 1 \mid e(X)\}.$$

The left-hand side of the equation equals:

$$\begin{aligned}\Pr\{Z = 1 \mid Y(1), Y(0), e(X)\} &= \mathbb{E}[Z \mid Y(1), Y(0), e(X)] \\ &= \mathbb{E}[\mathbb{E}[Z \mid Y(1), Y(0), e(X), X] \mid Y(1), Y(0), e(X)] \quad (\because \text{LIE.}) \\ &= \mathbb{E}[\underbrace{\mathbb{E}[Z \mid X]}_{=e(X)} \mid Y(1), Y(0), e(X)] = e(X).\end{aligned}$$

The right-hand side of the equation equals:

$$\Pr\{Z = 1 \mid e(X)\} = \mathbb{E}[Z \mid e(X)] = \mathbb{E}[\mathbb{E}[Z \mid e(X), X] \mid e(X)] \quad (\because \text{LIE.}) = \mathbb{E}[\underbrace{\mathbb{E}[Z \mid X]}_{=e(X)} \mid e(X)] = e(X)$$

□

11.1 The propensity score as a dimension reduction tool

- Thm. 11.1 motivaes *propensity score stratification*:

$$Z \perp\!\!\!\perp \{Y(1), Y(0)\} \mid e(X) = e_k \quad (k = 1, \ldots, K),$$

  where the propensity score only takes K posible values $\{e_1, \ldots, e_K\}$.
- In general, the propensity score is not known and is not discrete.
- We often fit a statistical model for $\Pr\{Z = 1 \mid X\}$ to obtain $\hat{e}(X)$ and we can discretize it to approximate the simple case above.
- For example, we can discretize $\hat{e}(X)$ by its K quantiles to obtain $\hat{e}'(X)$: $\hat{e}'(X) = e_k$, the k/K-th quantile of $\hat{e}(X)$, if $\hat{e}(X)$ is between the $(k-1)/K$-th and k/K-th quantitles of the $\hat{e}(X)$'s.

11.1 The propensity score as a dimension reduction tool

- Its rigorous quantification is still missing in the literature.
- Q1: How do we choose K?
  1. If K is too small, ignorability does not hold conditional on $\hat{e}'(X)$ even approximately.
  2. If K is too large, then we do not have enough units within each stratum of the estimated propensity score and many strata have only treated or control units.
- Q2: How do we compute standard errors of the estimators based on propensity score stratification?
  1. Reporting standard errors based on the SRE ignores the uncertainty in the estimated propensity scores.
  2. Some researchers employ the bootstrap method but the theory for the bootstrap is still unclear due to the discreteness of this estimator.

11.2 Propensity score weighting

Theorem 11.2

If $Z \perp\!\!\!\perp \{Y(1), Y(0)\} \mid X$ and $0 < e(X) < 1$, then

$$\mathbb{E}[Y(1)] = \mathbb{E}\left[\frac{ZY}{e(X)}\right], \quad \mathbb{E}[Y(0)] = \mathbb{E}\left[\frac{(1-Z)Y}{1-e(X)}\right].$$

- The additional assumption $0 < e(X) < 1$ is called *overlap* or *positivity* condition.
- If $e(X) = 1$ or $e(X) = 0$, Thm. 11.2 is not well defined.
- Moreover, if $e(X_i) = 1$, then it implies that a potential outcome $Y_i(0)$ does not exist, leading to a philosophical issue of causal inference.
- King and Zeng (2006) called $Y_i(0)$ an *extreme counterfactuals* when $e(X_i) = 1$.

Proof.

$$
\begin{aligned}
\mathbb{E}\left[\frac{ZY}{e(X)}\right] &= \mathbb{E}\left[\frac{ZY(1)}{e(X)}\right] = \mathbb{E}\left[\mathbb{E}\left[\frac{ZY(1)}{e(X)} \mid X\right]\right] \qquad (\because \text{LIE.}) \\
&= \mathbb{E}\left[\frac{1}{e(X)}\mathbb{E}\left[ZY(1) \mid X\right]\right] \\
&= \mathbb{E}\left[\frac{1}{e(X)} Y(1) \underbrace{\mathbb{E}[Z \mid X]}_{=e(X)}\right] \qquad (\because \text{strong ignorability.}) \\
&= \mathbb{E}[Y(1)]
\end{aligned}
$$

□

11.2 Propensity score weighting

- Thm. 11.2 motivates the following moment estimator for the average causal effect:

$$\hat{\tau}^{ht} = \frac{1}{n}\sum_{i=1}^{n}\frac{Z_i Y_i}{\hat{e}(X_i)} - \frac{1}{n}\sum_{i=1}^{n}\frac{(1-Z_i)Y_i}{1-\hat{e}(X_i)}.$$

- Recall that it is ideal to estimate $\tau = \mathbb{E}[Y(1) - Y(0)]$, but $Y(1), Y(0)$ are not simultaneously observable (the fundamental problem of causal inference).
- Thm. 11.2 enables us to estimate both $\mathbb{E}[Y(1)], \mathbb{E}[Y(0)]$.
- This is the *inverse propensity score weighting* (IPW) estimator, which is also called the Horvitz-Thompson (HT) estimator (Horvitz and Thompson, 1952).
- However, it is known that HT estimator has many problems.

Proposition 11.1 (lack of invariance for the HT estimator)

If we change Y_i to $Y_i + c$ with a constant c, then the HT estimator $\hat{\tau}^{ht}$ becomes $\hat{\tau}^{ht} + c(\hat{1}_T - \hat{1}_C)$, where

$$\hat{1}_T = \frac{1}{n}\sum_{i=1}^{n} \frac{Z_i}{\hat{e}(X_i)}, \quad \hat{1}_C = \frac{1}{n}\sum_{i=1}^{n} \frac{(1 - Z_i)}{1 - \hat{e}(X_i)}$$

can be viewed as two different estimates of the constant 1.

- With the true propensity scores, both $\hat{1}_T$ and $\hat{1}_C$ have expectation 1.
- In general, $\hat{1}_T - \hat{1}_C$ is not 0 in finite sample.

Proof.

We first prove Prop. 11.1. Substituting $Y_i + c$,

$$\frac{1}{n}\sum_{i=1}^{n}\frac{Z_i(Y_i+c)}{\hat{e}(X_i)} - \frac{1}{n}\sum_{i=1}^{n}\frac{(1-Z_i)(Y_i+c)}{1-\hat{e}(X_i)} = \hat{\tau}^{ht} + c\left(\underbrace{\frac{1}{n}\sum_{i=1}^{n}\frac{Z_i}{\hat{e}(X_i)}}_{=\hat{1}_T} - \underbrace{\frac{1}{n}\sum_{i=1}^{n}\frac{1-Z_i}{1-\hat{e}(X_i)}}_{=\hat{1}_C}\right).$$

We then prove that $\hat{1}_T$ has expectation 1 with the true propensity score.

$$\mathbb{E}[\hat{1}_T] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[\frac{Z_i}{e(X_i)}\right] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[\mathbb{E}\left[\frac{Z_i}{e(X_i)} \mid X_i\right]\right] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[\frac{1}{e(X_i)}\underbrace{\mathbb{E}\left[Z_i \mid X_i\right]}_{=e(X_i)}\right] = 1.$$

11.2 Propensity score weighting

- Hájek (1971) proposed a Hajek estimator:

$$\hat{\tau}^{\text{hajek}} = \frac{\sum_{i=1}^{n} \frac{Z_i Y_i}{\hat{e}(X_i)}}{\sum_{i=1}^{n} \frac{Z_i}{\hat{e}(X_i)}} - \frac{\frac{\sum_{i=1}^{n} (1-Z_i) Y_i}{1-\hat{e}(X_i)}}{\sum_{i=1}^{n} \frac{1-Z_i}{1-\hat{e}(X_i)}}.$$

- We can verify that if we replace Y_i by $Y_i + c$, then $\hat{\tau}^{\text{hajek}}$ remains the same.
- Many numerical studeis have found that $\hat{\tau}^{\text{hajek}}$ is much more stable than $\hat{\tau}^{ht}$ in finite samples.

11.2 Propensity score weighting

- Many asymptotic analyses require a *strong overlap* condition:

$$0 < \alpha_L \leq e(X) \leq \alpha_U < 1,$$

  that is, the true propensity score is bounded away from 0 and 1.
- This is a rather strong assumption, especially with many covariates.
- Even if this condition holds with the true propensity score, the estimated propensity score can be close to 0 or 1.
- When this happens, the weighting estimators blow up to infinity, which results in extremely unstable behavior in finite samples.

11.2 Propensity score weighting

- We can either truncate the estimated propensity score by changing it to

$$\max\left[\alpha_L, \min\left\{\hat{e}(X_i), \alpha_U\right\}\right],$$

  or trim the observations by dropping units with $\hat{e}(X_i)$ outside the interval $[\alpha_L, \alpha_U]$.
- Although trimming often stabilizes the IPW estimators, it also injects additional arbitrariness into the procedure.

11.3 The balancing property of the propensity score

Theorem 11.3

The propensity score satisfies $Z \perp\!\!\!\perp X \mid e(X)$. Moreover, for any function $h(\cdot)$, we have

$$\mathbb{E}\left[\frac{Zh(X)}{e(X)}\right] = \mathbb{E}\left[\frac{(1-Z)h(X)}{1-e(X)}\right]$$

provided the existence of the moments on both sides of the inequality.

- Thm. 11.3 does not require the ignorability assumption.
- The first part of Thm. 11.3 states that the covariate distributions are balanced across the treatment and control groups, within the same level of the propensity score.
- The second part of Thm. 11.3 states that an equivalent form of covariate balance based on the weighting form.

11.3 The balancing property of the propensity score

Proof.

1. The first statement is equivalent to $\Pr\{Z = 1 \mid X, e(X)\} = \Pr\{Z = 1 \mid e(X)\}$. Then, the left-hand side is

$$\Pr\{Z = 1 \mid X, e(X)\} = \Pr\{Z = 1 \mid X\} = e(X),$$

because $e(X)$ is a function of X so conditioning on X and $e(X)$ is same to conditioning on X. The right-hand side is

$$\begin{aligned}\Pr\{Z = 1 \mid e(X)\} &= \mathbb{E}[\mathbb{E}[Z = 1 \mid X, e(X)] \mid e(X)] = \mathbb{E}[\mathbb{E}[Z \mid X] \mid e(X)] \\ &= \mathbb{E}[e(X) \mid e(X)] = e(X).\end{aligned}$$

2. By LIE, we obtain the second part.

11.3 The balancing property of the propensity score

- Thm. 11.3 has useful implications for the statistical analysis: before getting access to the outcome data, we can check whether the propensity score model is specified well enough to ensure the covariate balance in the data.
- In propensity score weighting, we can view $h(X)$ as a pseudo outcome and estimate the average causal effect on $h(X)$.
- Because the true average causal effect on $H(X)$ is 0, the estimate should not be significantly different from 0. A canonical choice of $h(X)$ is X.

References I

Hájek, J. (1971). Comment on "an essay on the logical foundations of survey sampling, part one". *The foundations of survey sampling 236*.

Horvitz, D. G. and D. J. Thompson (1952). A generalization of sampling without replacement from a finite universe. *Journal of the American statistical Association 47*(260), 663–685.

King, G. and L. Zeng (2006). The dangers of extreme counterfactuals. *Political analysis 14*(2), 131–159.

Rosenbaum, P. R. and D. B. Rubin (1983). The central role of the propensity score in observational studies for causal effects. *Biometrika 70*(1), 41–55.